

External Verification Architecture (EVA)

The Missing Foundation for AI Safety:
Structure, Standardization,
and Implementation

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Concept and Terminology Notice

The concept of EVA (External Verification Architecture), its formal definition, and its three structural conditions — Verifiability, Transparency, and Physical Immutability — were proposed by Satoru Hara in 2026. This work is registered with a Digital Object Identifier (DOI) on Zenodo and SSRN.

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To all those who believe that
safety must be proved, not merely claimed.

Preface

This book emerges from an attempt to answer a question that has long remained unanswered in the international standardization of AI safety.

'AI must be safe' — this assertion appears repeatedly in the EU AI Act, ISO/IEC 42001, the NIST AI Risk Management Framework, and countless other international regulations and standards. Yet one critical question has gone unanswered:

How do we technically prove that an AI system is safe?

In response, the author proposes External Verification Architecture (EVA) — a technical framework for continuously, structurally, and verifiably monitoring, recording, and evaluating the reasoning processes, outputs, and decision-making structures of AI systems through mechanisms independent of the model's internal parameters.

What EVA addresses is not a problem of capability. It is a problem of structure. Current AI systems are not lacking in ability. What is lacking, by design, is a mechanism for externally proving that those abilities are being used appropriately.

The analogy is precise: the aviation industry did not make aircraft safer by building more powerful engines. It made aircraft safer by mandating flight recorders, standardizing maintenance protocols, and requiring independent accident investigations. EVA proposes the same structural shift for AI.

This book begins with EVA's conceptual definition, proceeds through its technical implementation pathways and international standardization roadmap, and concludes with an examination of the Human Natural Structure (HNS) framework as one technical method for realizing EVA.

It is the author's hope that this work will accelerate the transition in AI safety assurance — from assertion to proof.

May 2026 — Satoru Hara

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Chapter 1 The Limitations of Current AI Safety Assurance

As AI systems are rapidly deployed across society, there is growing international demand for frameworks that objectively guarantee their safety, reliability, and ethical compliance. Yet in practice, AI safety assurance is predominantly dependent on internal evaluations designed and conducted by the model developers themselves — and no established mechanism exists for independent third-party technical verification.

This chapter examines the structural nature of this gap, why it exists, and why existing approaches cannot close it.

1.1 The Structural Problem of Self-Assessment Dependency

The mainstream of current AI safety evaluation consists of internal assessments designed and conducted by model developers. Representative methods include Red Teaming, RLHF (Reinforcement Learning from Human Feedback), Constitutional AI, and internal benchmark evaluation — all of which have produced meaningful results.

However, these methods share a common structural limitation: the design, execution, and interpretation of evaluation results all fall within the discretion of the same developer, with no institutional guarantee of independent third-party verification. This represents a clear institutional gap when compared to the independence requirements of clinical trials in pharmaceuticals or external audits in accounting.

In the pharmaceutical industry, a drug manufacturer cannot self-certify the safety of its own products — an independent clinical trial is mandatory. In financial accounting, a company cannot audit its own books. AI safety currently has no equivalent structure.

No matter how sincere the model developer, self-assessment carries a structural tendency to overlook risks that the developer themselves has not anticipated.

1.2 The Absence of Technical Verifiability

A key technical barrier to external verification is the opacity of AI reasoning processes. Modern large language models constitute a black box from which it is extremely difficult to causally explain externally why a particular output was generated.

For an external verification mechanism to render judgments, structured data must be generated in real time, in parallel with inference. Current AI architectures are not designed with the assumption of connectivity to such external verification mechanisms. This is not a technical accident — it is the result of a design choice. EVA aims to fill this design-level gap.

The absence of technical verifiability means that even when an AI system produces a harmful output, it is often impossible to determine: which step led to the error, whether the error was systematic or isolated, and whether any corrective measure was actually effective.

1.3 The Gap Between Regulatory Requirements and Technical Implementation

The EU AI Act mandates log-keeping and human oversight for high-risk AI systems. However, the technical specifications for log-keeping are not stipulated: what to record, at what granularity, and in what format are left entirely to the developer's interpretation.

ISO/IEC 42001:2023 provides a framework for AI management systems, but its scope of certification covers processes, not independent technical verification of reasoning processes themselves.

Closing the gap between regulatory requirements and technical implementation is EVA's central mission.

1.4 Three Real-World Consequences

The absence of external verification architecture produces three concrete consequences.

First, undetectable systematic errors. When an AI system contains a systematic bias in its reasoning, this error can persist indefinitely because no external mechanism exists to detect it at the inference level.

Second, ineffective corrective measures. When a problem is identified and developers implement corrections, it is impossible without EVA to

confirm whether the correction addressed the root cause or merely suppressed the visible symptom.

Third, unverifiable safety claims. AI developers regularly make public statements about the safety of their systems. Without external verification infrastructure, these statements cannot be independently validated.

Chapter 2 Conceptual Definition of EVA

2.1 What is EVA?

External Verification Architecture (EVA) is a technical framework for continuously, structurally, and verifiably monitoring, recording, and evaluating the reasoning processes, outputs, and decision-making structures of AI systems through external mechanisms independent of the model's internal parameters.

The essence of EVA lies in the structural guarantee of externality. As long as the verification mechanism depends on the same developer, the same software environment, or the same hardware substrate as the model itself, that verification is effectively no different from self-assessment.

EVA is not a specific product, algorithm, or service. It is an architectural concept — a definition of what properties a verification system must have to qualify as genuinely external.

2.2 The Three Structural Conditions

For EVA to function effectively, the following three conditions must be satisfied simultaneously.

Condition	Definition	Technical Implication
Verifiability	Each reasoning step is output externally in machine-readable format	Structured logs in JSON-LD / RDF. Third parties must be able to analyze programmatically
Transparency	A monitoring mechanism operates in parallel with inference	Operates independently of model inference. The model cannot manipulate the monitoring mechanism
Physical Immutability	Verification logic is on a substrate that cannot be altered by the model	Embedded in ROM / FPGA / ASIC. Inaccessible via software updates

These three conditions constitute the logical minimum requirements. Verifiability alone cannot eliminate the risk of logs being falsified. Transparency alone leaves the possibility that the model circumvents the mechanism. Physical Immutability alone allows the verification criteria to

become arbitrary. Only when all three conditions overlap does genuine external verification become possible.

2.3 What EVA Achieves: From Assertion to Proof

The fundamental change EVA achieves is a shift in AI safety assurance from assertion to proof. Without EVA, a developer states: 'Our model has been extensively tested and meets all applicable safety requirements.' This cannot be independently verified.

With EVA, the developer can state: 'Our model's reasoning processes are continuously recorded in JSON-LD format to an immutable external log, which any accredited third party can inspect.' This is a verifiable claim.

This shift — from assertion to proof — is precisely what occurred in aviation safety, pharmaceutical regulation, and financial auditing. EVA proposes the same structural transformation for AI.

2.4 EVA as a Blueprint for Post-Incident Corrective Action

The value of EVA extends beyond static proof. It also functions as a structural blueprint for corrective action when problems occur.

Without EVA: the cause of a harmful output cannot be identified; the reasoning step where the error occurred cannot be traced; and whether any fix was effective cannot be confirmed.

With EVA: a problem occurs → structured logs are examined to identify the exact reasoning coordinate of the deviation → a targeted intervention is designed → subsequent logs confirm the deviation no longer occurs.

EVA is not only a structure for statically proving safety — it is a structural foundation for continuously improving AI systems through evidence-based corrective cycles.

2.5 How EVA Differs from Existing Approaches

Approach	Focus	EVA Difference
XAI	Post-hoc interpretation	EVA is real-time, not post-hoc
AI Auditing	Periodic snapshot	EVA is continuous — every

	evaluation	inference
Model Cards	Documentation of design intent	EVA verifies actual runtime behavior
ISO/IEC 42001	Process certification	EVA verifies inference-level outcomes

Chapter 3 Technical Implementation of EVA

3.1 The Two-Layer Structure

Full implementation of EVA is achieved through the integration of two layers: the software layer and the hardware layer.

Layer	Conditions Satisfied	Key Technologies
Software	Verifiability, Transparency	Sidecar, JSON-LD, W3C PROV-O
Hardware	Physical Immutability	FPGA / ASIC, TPM, Secure Enclave

3.2 Software Layer: Sidecar Architecture

EVA's software layer is implemented as a sidecar — a separate process running alongside the main AI inference process, with no shared memory that the model can access or manipulate.

The sidecar writes the following as structured logs at each step of inference:

- **Structural classification of inputs:** coordinate assignment within the HNS-36 reference baseline structure
- **Intermediate representation tracking:** time-series recording of reasoning position at each step
- **Output provenance mapping:** traceable chain from input structure to output
- **Structural deviation detection:** automatic flagging of anomalous reasoning flows

Logs are output in JSON-LD format, compliant with the W3C PROV-O ontology, in a state analyzable by independent third parties. The model body and log buffer operate in strictly separated process spaces.

3.3 Hardware Layer: Physical Immutability

The highest-standard EVA requires physical immutability at the hardware layer to eliminate the risk of the verification system itself being overwritten.

- **FPGA:** Verification logic burned in at manufacture. Unmodifiable by software updates.
- **ASIC:** Dedicated chip. Highest tamper resistance for large-scale deployment.
- **TPM:** Cryptographic signing of logs. Hardware-rooted integrity guarantee.
- **Secure Enclave (Intel SGX, ARM TrustZone):** Hardware-isolated execution, protected from the host OS.

3.4 Implementation by Sector

Sector	Key Risk	Recommended Approach
Healthcare	Misdiagnosis	Software sidecar + TPM integrity. Every clinical recommendation logged.
Finance	Biased decisions	FPGA-anchored verification + real-time anomaly detection.
Government	Discriminatory outcomes	ASIC implementation. Independent third-party log access mandated.
Infrastructure	Safety-critical failures	Hardware-embedded EVA in edge devices. Cryptographic log synchronization.

Chapter 4 International Standardization of EVA

4.1 Why Standardization is Indispensable

As long as EVA remains a proprietary implementation by individual AI developers, its social impact will be limited. For EVA to function as infrastructure for proving AI safety, it must be standardized internationally.

First, interoperability. Without a common standard, AI systems from different vendors will produce EVA logs in incompatible formats, making cross-vendor third-party verification impossible.

Second, regulatory effectiveness. The EU AI Act mandates external verification, but the technical specifications remain undefined. A standardized EVA specification translates regulatory language into engineering requirements.

Third, transferability of trust. EVA's credibility must rest on the authority of independent international standards bodies — ISO, IEC, IEEE, ITU — not on the reputation of any individual company.

4.2 Alignment with Major Standards Bodies

Body	EVA Proposal
ISO/IEC JTC 1/SC 42	New Work Item Proposal: Technical Specifications for External Verification Architecture of AI Systems
CEN-CENELEC JTC 21	Incorporate EVA into EN for EU AI Act Articles 12 and 13
NIST AI RMF	EVA as technical implementation of the MEASURE function
IEEE SA	EVA as technical verification method for AI transparency (IEEE P2863)
ITU-T AI	EVA as external verification requirements for AI in telecommunications infrastructure

4.3 Concrete Pathway to Standardization

- **Stage 1:** Establish public technical documentation — Publish EVA specifications as DOI-assigned preprints on Zenodo and SSRN. This book is part of that effort.
- **Stage 2:** Technical proposal to national standards bodies — Submit EVA as NWI to ISO/IEC SC 42 through Japan's JISC / ITSCJ.
- **Stage 3:** Collaboration with European standards bodies — Propose EVA to CEN-CENELEC JTC 21 for EU AI Act alignment.
- **Stage 4:** Build implementation track record — Conduct PoC implementations with AI companies to generate empirical evidence.
- **Stage 5:** Transfer to independent foundation — Transfer EVA governance to a multi-stakeholder foundation to ensure neutrality and sustainability.

4.4 Social Impact of Standardization

- EVA conformance will be specified as a technical requirement in AI procurement across government, healthcare, finance, and infrastructure
- A third-party AI safety certification market will emerge
- Submitting proof of safety rather than assertions will become standard practice
- EVA conformance will become a basis for mutual recognition in international AI trade
- AI liability insurance products will be more precisely structured using EVA behavioral data

Chapter 5 Human Natural Structure (HNS)

This chapter introduces the Human Natural Structure (HNS) framework as one technical method for realizing EVA. HNS is one candidate reference baseline; its full technical specification is available on GitHub.

5.1 EVA Requires a Reference Baseline

For EVA to function as more than a recording mechanism, the verification process must answer: what criteria are used to judge whether AI reasoning is structurally appropriate?

This reference baseline must be: culturally and linguistically neutral; applicable across domains; defined in machine-readable format; and independent of developer subjectivity. No such baseline has been established in current AI research or international standards.

5.2 HNS as a Reference Baseline for EVA

The Human Natural Structure (HNS) framework consists of four architectural layers:

- **HumanOS** — Conceptual integration layer (blueprint and meta-architecture)
- **HNS-36** — Foundational Specification: the structural coordinate system
- **HNS-144** — Observational Specification: the observational operating system
- **HNS-864** — Analytical Specification: the analytical operating system

For EVA's purposes, HNS-36 provides the primary reference baseline: a 36-cell structural coordinate system formed as the product of six Human Natural Layers and six Abstract Cognitive Categories, describing human cognition, behavior, and social meaning in a culturally invariant, formally specifiable form.

Component	Content
6 Human Natural Layers	PhysicalOS / CognitiveOS / InteractionOS / EnvironmentOS / LoadOS / PatternOS

6 Abstract Cognitive Categories	Civilization / Core / Module / Application / System / External
36-Cell System	6 layers $\times$ 6 categories = 36 structural coordinates for mapping AI reasoning to human cognitive structure

This coordinate system is defined in JSON-LD / RDF. In relation to EVA's conditions: JSON-LD logs satisfy Verifiability; sidecar-based assignment satisfies Transparency; FPGA / ASIC embedding satisfies Physical Immutability.

5.3 The Positioning of HNS within EVA

EVA defines the requirements; HNS is one method for satisfying them. The EVA standard should define the properties any reference baseline must have — HNS satisfies those properties, but other architectures can in principle do so as well.

HNS is one technical method for realizing EVA.

HNS is publicly available at https://github.com/satoru-hara/03_NSW. For comprehensive technical detail, see the HNS Specification Series (Hara, 2026).

Column: EVA as the Flight Recorder for AI

On July 11, 1962, a United Airlines Boeing 707 crashed near Uniontown, Pennsylvania, killing all 45 people on board. Investigators were largely unable to determine the precise cause. The flight data recorder was of limited capability, and the cockpit voice recorder did not yet exist as a mandatory requirement.

Eleven years later, mandatory cockpit voice recorders became standard on commercial aircraft worldwide. Today, modern aircraft record hundreds of parameters continuously, with that data retained in tamper-resistant hardware designed to survive catastrophic accidents.

The result was transformative. The rate of fatal accidents per billion passenger-kilometers fell by more than 95% over the following four decades. Not because aircraft became different machines. Not because pilots became different humans. But because the industry gained the ability to identify the precise cause of each accident and translate that precision into targeted, verifiable corrective action.

AI systems in 2026 are where commercial aviation was in 1965.

The systems are powerful. The capabilities are impressive. And when something goes wrong, we largely cannot determine why, cannot verify whether our fix addressed the root cause, and cannot confirm that the same failure will not recur.

EVA proposes, in essence, the flight recorder mandate for AI — not as a regulatory burden, but as the structural prerequisite for systematic safety improvement.

The question is not whether AI needs external verification architecture. The question is only how soon we will build it.

The answer will be determined, in large part, by whether the international standards community moves to standardize EVA before the next generation of high-stakes AI deployments makes the absence of such architecture impossible to ignore.

This book is a contribution to that conversation.

Appendix A NWI Draft

Draft New Work Item Proposal for EVA standardization in ISO/IEC JTC 1/SC 42, prepared for submission via JISC / ITSCJ.

Item	Content
Proposed Title	Information technology — Artificial intelligence — Technical Specification for External Verification Architecture (EVA) of AI Systems
Proposing Body	Japan (JISC: Japanese Industrial Standards Committee)
Proposer	Satoru Hara / Natural Structure Works (NSW)
Submission Route	Via Information Technology Standards Commission of Japan (ITSCJ)
Document Type	Technical Specification (TS) → International Standard (IS)
Scope	EVA definition, three structural conditions, implementation requirements, conformance criteria, alignment with ISO/IEC AI standards
Target Schedule	NWI: 2026 / WD 1.0: H1 2027 / IS/TS: H2 2028
Related Standards	ISO/IEC 42001 / 23894 / 22989 / EU AI Act (2024/1689) / NIST AI RMF 1.0
Cooperating Bodies	CEN-CENELEC JTC 21 / IEEE SA / ITU-T AI Focus Group / NIST
Declaration	Independent researcher with no conflict of interest with any company or institution.

Appendix B References

- [1] European Parliament and Council. (2024). Regulation (EU) 2024/1689 — Artificial Intelligence Act.
- [2] ISO/IEC. (2023). ISO/IEC 42001:2023 — AI Management system.
- [3] ISO/IEC. (2023). ISO/IEC 23894:2023 — AI Risk management.
- [4] ISO/IEC. (2022). ISO/IEC 22989:2022 — AI Concepts and terminology.
- [5] ISO/IEC. (2022). ISO/IEC 23053:2022 — Framework for AI Systems Using ML.
- [6] ISO/IEC. (2020). ISO/IEC TR 24028:2020 — Trustworthiness in AI.
- [7] NIST. (2023). AI Risk Management Framework (AI RMF 1.0). <https://doi.org/10.6028/NIST.AI.100-1>
- [8] ENISA. (2020). AI Cybersecurity Challenges. EU Agency for Cybersecurity.
- [9] W3C. (2013). PROV-O: The PROV Ontology. <https://www.w3.org/TR/prov-o/>
- [10] Trusted Computing Group. (2019). TPM Library Specification, Family 2.0.
- [11a] Hara, S. (2026). HumanOS: Human Natural Structure Operating System. Natural Structure Works (NSW). https://github.com/satoru-hara/03_NSW
- [11b] Hara, S. (2026). HNS-36: Human Natural Structure Foundational Specification (v1.1). Natural Structure Works (NSW). https://github.com/satoru-hara/03_NSW
- [11c] Hara, S. (2026). HNS-144: Human Natural Structure Observational Specification (v1.1). Natural Structure Works (NSW). https://github.com/satoru-hara/03_NSW
- [11d] Hara, S. (2026). HNS-864: Human Natural Structure Analytical Specification (v1.1). Natural Structure Works (NSW). https://github.com/satoru-hara/03_NSW

- [12] Hara, S. (2026). EVA: Structural Significance and Importance in AI. [Zenodo / SSRN DOI]
- [13] LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, 521, 436–444.
- [14] Russell, S., & Norvig, P. (2021). *Artificial Intelligence: A Modern Approach* (4th ed.). Pearson.
- [15] Varela, F. J., Thompson, E., & Rosch, E. (1991). *The Embodied Mind*. MIT Press.

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with active engagement with ISO/IEC JTC 1/SC 42, CEN-CENELEC JTC
21, JISC, ITSCJ, and METI.

Key activities:

- EVA (External Verification Architecture): concept proposed and registered on Zenodo / SSRN with DOI — May 2026
- HNS (Human Natural Structure): four-specification series publicly available on GitHub (HumanOS, HNS-36, HNS-144, HNS-864)
- NWI draft prepared for ISO/IEC JTC 1/SC 42 submission
- Technical proposal submitted to CEN-CENELEC JTC 21
- Standardization activities with JISC / ITSCJ / METI (Japan)

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